

Dashboard / Meine Kurse / 2023S365249 / Final Exam 29th June -- 17:15 -- 21:00 / [Exam] PG 1

Begonnen am Thursday, 20. April 2023, 17:09

Status Beendet

Beendet am Thursday, 20. April 2023, 17:24

Verbrauchte Zeit 14 Minuten 59 Sekunden

Punkte 6,00/9,00

Bewertung 6,67 von 10,00 (66,67%)

Frage 1

Vollständig

Erreichte Punkte 1,00 von 1,00

We call $J(\pi)$ the performance of a policy π . What do we mean with $J(\pi)$? ($R(\tau)$ represents the return for a trajectory τ)

Wählen Sie eine oder mehrere Antworten:

- ☒ a. $J(\pi) = E_{\tau \sim p_{\pi}(\tau)}[R(\tau)]$
- ☐ b. $J(\pi) = \min_{\pi} E_{\tau \sim p_{\pi}(\tau)}[\sum_{t=0}^T R_{t+1}]$
- ☐ c. $J(\pi) = \max_{\pi} E_{\tau \sim p_{\pi}(\tau)}[R(\tau)]$
- ☐ d. $J(\pi) = \max_{\pi} E_{\tau \sim p_{\pi}(\tau)}[\sum_{t=0}^T R_{t+1}]$
- ☒ e. $J(\pi) = E_{\tau \sim p_{\pi}(\tau)}[\sum_{t=0}^T R_{t+1}]$
- ☐ f. $J(\pi) = \min_{\pi} E_{\tau \sim p_{\pi}(\tau)}[R(\tau)]$

Frage 2

Vollständig

Erreichte Punkte 1,00 von 1,00

Why computing the gradient $\nabla_{\theta} J(\theta_t)$ is tricky?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. because $J(\theta)$ depends on both the action selection (directly determined by π_{θ}) and the stationary distribution of states following the target selection behavior (indirectly determined by π_{θ})
- ☐ b. because $J(\theta)$ uses a *min* operator in its formulation, which is not differentiable. Therefore a proxy must be used instead.
- ☐ c. because $J(\theta)$ uses a *max* operator in its formulation, which is not differentiable. Therefore a proxy must be used instead.

Frage 3

Vollständig

Erreichte Punkte 0,00 von 1,00

Which of the following is an advantage of using the Policy Gradient algorithm over value-based methods?

- ☒ a. Policy Gradients can handle large state spaces more efficiently than value-based methods.
- ☐ b. Policy Gradients do not require a model of the environment.
- ☐ c. Policy Gradients can learn optimal policies for both continuous and discrete action spaces.
- ☐ d. Policy Gradients can learn stochastic policies.

Frage 4

Vollständig

Erreichte Punkte 1,00 von 1,00

What is the intuition of policy gradient theorem?

$$\nabla_{\theta} E_{\tau \sim p_{\pi_{\theta}}(\tau)} [R(\tau)] = E_{\tau \sim p_{\pi_{\theta}}} [R(\tau) \nabla_{\theta} \log \pi_{\theta}(\tau)]$$

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The term $\sum_{t=0}^T \log \pi_{\theta}(a_{i,t} \mid s_{i,t})$ measures the likelihood of the current trajectory given the policy π_{θ} .
- ☒ b. The term $\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_{i,t} \mid s_{i,t})$ is the gradient of the log likelihood.
- ☐ c. The gradient $\nabla_{\theta} E_{\pi_{\theta}} [R(\tau)]$ increases the likelihood of trajectories with low returns.
- ☒ d. The gradient $\nabla_{\theta} E_{\pi_{\theta}} [R(\tau)]$ increases the likelihood of high return trajectories
- ☒ e. The gradient $\nabla_{\theta} E_{\pi_{\theta}} [R(\tau)]$ decreases the likelihood of trajectories with low returns.
- ☐ f. The gradient $\nabla_{\theta} E_{\pi_{\theta}} [R(\tau)]$ decreases the likelihood of high return trajectories

Frage 5

Vollständig

Erreichte Punkte 0,33 von 1,00

What of the following statements are true when referred to the main problems of on-policy policy gradient?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Past experiences generated with old policies cannot be used to approximate the true gradient
- ☒ b. It is very sample inefficient
- ☐ c. Expert trajectories cannot be used but past experiences with old policies can be used to approximate the true gradient
- ☐ d. Experiences coming from a different policy cannot be used to approximate the true gradient

Frage 6

Vollständig

Erreichte Punkte 0,50 von 1,00

What does Importance Sampling allow us to do when computing the approximation of the true gradient in policy gradient?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. It allows us to get a better approximation of the true gradient
- ☐ b. It allows us to compute the true gradient instead of an approximation
- ☒ c. It allows us to compute the expected value of one random variable which follows one distribution, by sampling it with a different distribution
- ☐ d. It allows us to use trajectories from different policies to approximate the true gradient

Frage 7

Vollständig

Erreichte Punkte 0,67 von 1,00

What of the following statements is/are true when referred to baseline subtraction on policy gradient.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Subtracting a baseline is biased in expectation, as long as the baseline does not depend on actions
- ☐ b. The value function is the best choice for the baselines
- ☐ c. Subtracting a baseline is biased in expectation, as long as the baseline depends on actions
- ☒ d. The value function is a good choice for the baselines, but not the optimal
- ☐ e. Subtracting a baseline is unbiased in expectation, as long as the baseline depends on actions
- ☒ f. Subtracting a baseline is unbiased in expectation, as long as the baseline does not depend on actions

Frage 8

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following techniques is commonly used to reduce the variance of policy gradient estimates in reinforcement learning?

- ☒ a. Baseline subtraction
- ☐ b. Actor-Critic
- ☐ c. Q-Learning
- ☐ d. Importance Sampling

Frage 9

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following statements about the Policy Gradient algorithm is true?

- ☐ a. Policy Gradients require the agent to learn a state-value function.
- ☐ b. Policy Gradients are a type of model-free reinforcement learning algorithm.
- ☐ c. Policy Gradients are only applicable to continuous action spaces.
- ☒ d. Policy Gradients update the policy parameters in the direction of the gradient of the expected reward with respect to the policy parameters.

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Direkt zu:

[\[Exam\] PG 2 ▶](#)